

Outage Investigation Report —

OIR-2025-003

Field	Value
Document ID	OIR-2025-003
Report Date	2025-10-23
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Reviewing Manager	Priya Ramachandran, Plant Manager, NW-PowerPool
Region	NW-PowerPool
Site	NW-PowerPool-WIND-003
Asset	WIN-10085 (Vestas, wind_turbine)
Outage ID	OUT-2025-0476
Classification	Forced outage — overheating
Distribution	NW-PowerPool Operations, Reliability Engineering, Vendor Management, OEM Liaison

1. Outage Summary

WIN-10085 declared a forced outage on 2025-10-07 at 07:07 PDT after a gearbox over-temperature trip. Return to service occurred on 2025-10-16 at 17:07 PDT, total duration **226 hours**. Lost generation: **3,740.6 MWh**. This is the opening event of what Reliability Engineering has begun referring to as the **Fall Overheating Sequence** (October–December 2025), formally classified under RCA-2025-002.

WIN-10085 is a Vestas-supplied unit. Vestas fleet history at MISO-WIND-038 has been the lowest-failure baseline in the Cascadia portfolio, which makes this event notable as a deviation rather than a fleet trend. The trip mechanism is consistent with a maintenance-side cause rather than an OEM design issue.

2. Timeline of Events

Timestamp (PDT)	Event
2025-10-06 22:40	SCADA shows gearbox oil sump temperature trending upward from 71 °C to 79 °C over 6 hours. Ambient cooled overnight to 4 °C, so the heat rise is load-related rather than ambient.
2025-10-07 04:55	Sump temperature crosses Vestas OEM "monitor" threshold of 82 °C.
2025-10-07 07:07	Trip at 89.7 °C sump / 94.1 °C main bearing. Forced outage declared.
2025-10-07 09:30	CPS dispatched from Bend, OR depot — 4 h 20 min ETA. Under EPA-2024-11 the 4-hour threshold was marginally exceeded, but on-call manager elected to wait for CPS rather than activate Apex. (See EMR-2025-004 dissent memo.)
2025-10-07 13:50	CPS onsite. Initial inspection finds heat-exchanger fins fouled with airborne particulate from the August wildfire smoke episode; coolant flow reduced ~35%.
2025-10-08 — 2025-10-14	Heat exchanger pull-and-clean, gasket replacement, oil change, vibration baseline. Spectrometer reading 8 ppm iron (within limit).
2025-10-15 — 2025-10-16	Restart sequence and 24-h surveillance run.
2025-10-16 17:07	Returned to service.

3. Telemetry Observations

- Gearbox oil sump at trip: 89.7 °C.
- Main bearing at trip: 94.1 °C.
- Coolant flow rate at trip: 41 L/min (rated 64 L/min).
- Iron particulate in oil: 8 ppm — within limits, supporting the heat-exchanger fouling hypothesis.
- Vibration RMS: 4.2 mm/s (well below MS-2025-03 limit 6.5).
- Heat-exchanger fin spacing audit: 11 of 24 channels >50% blocked with fine particulate; consistent with the 2025-08-09 through 2025-08-21 regional wildfire smoke event (NW-PowerPool ambient AQI sustained >180).

The cooling system inspection scheduled for September under WO-2025-09081 was closed "complete, no findings" by Pinnacle Energy Maintenance. Photo evidence attached to that WO does not include heat-exchanger fin images. See CAR-2025-004 and CAR-2025-007.

4. Restoration Activities

Vendor	Scope	Invoice (USD)
Cascadia Power Services (CPS)	Heat-exchanger pull-and-clean, gasket replacement, oil change, vibration baseline	\$94,800
Vestas OEM (technical advisory remote)	Procedure review and warranty sign-off	\$11,200
Total		\$106,000

This is roughly 1/4 the Apex equivalent on WIN-10131 (OIR-2025-002, \$385,000), reinforcing the cost arbitrage analysis in EMR-2025-002 and RCA-2025-001.

5. Preliminary Root Cause

Direct cause: heat-exchanger fin fouling from wildfire smoke particulate, leading to reduced coolant flow and runaway sump temperature under sustained 7.4 MW load.

Contributing causes:

1. Cooling system inspection WO-2025-09081 was closed without verifying the heat-exchanger fins. Pinnacle Energy Maintenance documentation deficiency — see CAR-2025-004.
2. There is no written Cascadia procedure for post-wildfire-smoke inspection. The OEM manual (Vestas Service Procedure VSP-3.2) does not contemplate sustained particulate exposure; the gap should be addressed in CSM-7. See CAR-2025-007.

6. Recommendations

1. Issue a one-time inspection sweep of heat-exchanger fins at all NW-PowerPool wind assets within 30 days. Track under WO-2025-10444 (in preparation).
2. Add post-particulate-event inspection to CSM-7. Refer to CAR-2025-007.
3. Audit Pinnacle Energy Maintenance close-out documentation across NW-PowerPool-WIND-003 and -024 for the September 2025 PM cycle. Refer to CAR-2025-004.

4. Add WIN-10085 cooling system to FY26 routine refurbishment schedule. Internal estimate \$185K per turbine.
5. Cross-reference RCA-2025-002 for the broader Fall Overheating Sequence and OIR-2025-004 (TRA-10561, same site, December event).

Signatures

Role	Name	Date
Author	Marcus Chen, Sr. Reliability Engineer	2025-10-23
Plant Manager	Priya Ramachandran	2025-10-24
Vendor Manager	L. Halverson	2025-10-24

Distribution: NW-PowerPool Operations · Reliability Engineering · Vendor Management · OEM Liaison · CFO Office

References: OIR-2025-004 · CAR-2025-004 · CAR-2025-007 · EMR-2025-002 · EMR-2025-004 · RCA-2025-001 · RCA-2025-002 · Vestas Service Procedure VSP-3.2 · CSM-7 · Standard MS-2025-03